

The Mythos of Model Interpretability

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The Core Problem

- Supervised machine learning models boast remarkable predictive capabilities.
- But can we **trust** these models?
- Will they work reliably in deployment?
- “Model interpretability” is often proposed as a solution, but what does it actually mean?
- Many papers call models interpretable without defining what they mean.
- Lipton argues this makes interpretability claims vague and sometimes quasi-scientific.
- The paper tries to clarify both:
 - ▶ why interpretability is demanded
 - ▶ and what model properties are supposed to provide it.
- Common oversimplification: “linear models are interpretable, deep neural networks are not.” The paper argues this is too simplistic.

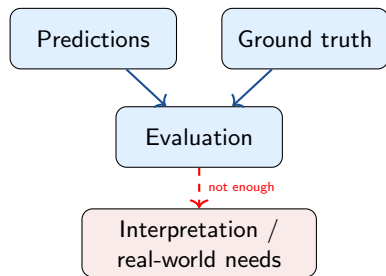
Main aim

Clarify:

- why interpretability is demanded
- what model properties are supposed to provide it

Desiderata of Interpretability Research

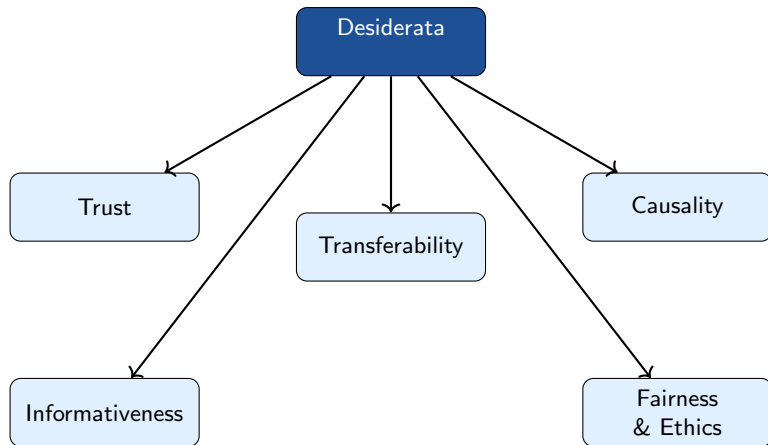
- Interpretability becomes important when there is a **mismatch** between:
 - ▶ Training objective (accuracy, loss)
 - ▶ Real-world deployment goals
- Predictions and evaluation metrics alone are often **insufficient**.
- The very desire for an interpretation implies that predictions alone do not characterize the model.
- Real-world goals are difficult to encode as simple mathematical functions.



Evaluation metrics are insufficient – hence we need interpretation.

Example

A hiring model may need to optimize productivity, ethics, and legality, but ethics and legality are hard to write as a simple loss function.

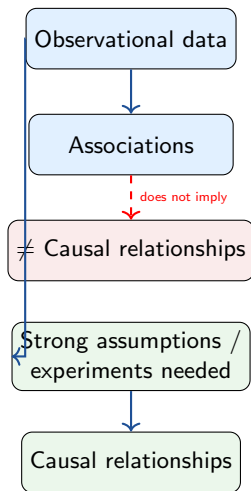


- Some papers say interpretability is needed for trust.
- But trust can mean different things:
 - ▶ confidence in performance (predictive accuracy)
 - ▶ subjective comfort with a well-understood model
 - ▶ confidence that the model behaves correctly against *real-world* objectives in deployment
- A model may also be considered trustworthy if it makes mistakes in the same regions where **humans also make mistakes**.
- We must align interpretability with real-world needs, not just model performance.

Example

An algorithm predicts crime rates to help a city allocate police officers. The model is highly accurate, but its training data contains racial biases. Trusting this “black box” blindly results in over-policing specific neighborhoods, perpetuating a cycle of incarceration. We cannot trust accuracy metrics alone; interpretability is needed to verify ethical behavior.

- Supervised learning models are only optimized to make **associations**, not causal inferences.
- But researchers often use them hoping to infer causal properties or generate hypotheses about the world.
- Associations are **not** guaranteed causal relationships.
- Author refers that some papers emphasize regression trees and Bayesian neural networks, suggesting such models can provide clues about causal relationships.
- We generally use observational data, these methods for inferring causality from observational data require strong prior assumptions.



- Train-test evaluation assumes future data will come from the same distribution.
- In deployment, that assumption can fail because environments are non-stationary.
- Adversarial settings are even more extreme . e.g., CNNs fooled by imperceptibly perturbed images ,these are mistakes a human would not make.
- Interpretability is desired to judge whether a model will behave sensibly outside its original setting.

Example 1

Product recommendation is non-stationary because new products appear and user preferences shift over time.

Example 2

A pneumonia risk model learned that asthma patients seemed lower-risk because they historically received more aggressive treatment. If deployed for triage, that treatment pattern would change, invalidating the model.

Example 3

FICO uses logistic regression partly for interpretability. But features like debt ratio or number of accounts can be strategically gamed by users without changing underlying ability to repay, weakening predictive validity.

- Sometimes the purpose of a model is not just to make a decision, but to give useful information to a human decision-maker.
- An interpretation can be useful even if it does not reveal the true internal mechanism of the model.
- **Informativeness is therefore distinct from transparency.**

Example

1. A diagnosis model may help by showing similar past cases that support its decision.
2. The paper uses the analogy of a PhD advisor: Suppose the student asks what venue would best suit a paper. The advisor could simply name one conference, but this may not be especially useful. the advisor should give information the student can combine with her own knowledge to make a decision.

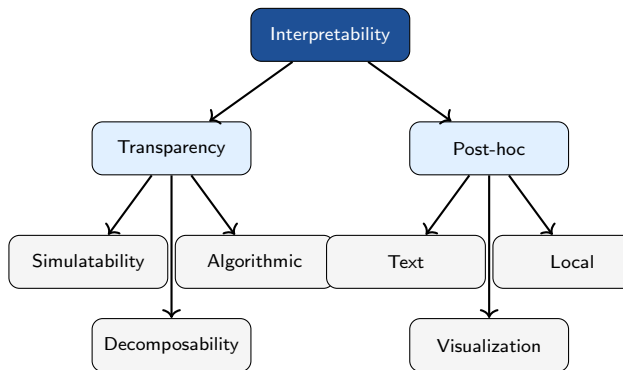
- Interpretability is often demanded to evaluate whether automated decisions are **fair and ethical**.
- This is critical in credit scoring, hiring, recidivism prediction, and news curation.
- Standard metrics like accuracy or AUC do **not** guarantee fair behavior.
- The EU “right to explanation” requires clear, falsifiable reasoning and a way to **contest** decisions – current interpretability methods mostly do not address contestability.

Example

Recidivism predictions are already used to determine who to release and who to detain, raising ethical concerns. How can we be sure that predictions do not discriminate on the basis of race?

Properties of Interpretable Models

- The paper divides interpretability into two large groups:
 - ▶ **Transparency** – how does the model work?
 - ▶ **Post-hoc interpretability** – what useful information can we extract after training?
- These are related, but they are not the same thing.
- Humans may be interpretable in a post-hoc sense (they can explain decisions), even though the brain itself is mechanistically opaque.



Transparency 1: Simulatability

- A model is simulatable if a human can grasp the whole model and step through the computation in a reasonable amount of time.
- This is why simple linear models and short decision trees are often called interpretable.
- But the paper argues this depends on **actual model size and complexity**, not just model family.

Example

Sparse linear models from LASSO are often treated as more interpretable than dense linear models. But a very large linear model, a very deep decision tree, or a long rule list may no longer be simulatable by a human.

Transparency 2: Decomposability

- A model is decomposable if each input, parameter, and computation has an **intuitive meaning**.
- This requires not only simple parameters, but also understandable features.
- Decomposability can fail when features are heavily engineered or anonymized.

Example

In a decision tree, a node like “patients with diastolic blood pressure over 150” is easy to explain. In a linear model, weights may seem meaningful, but the paper warns they can change with feature selection. For instance, the association between flu risk and vaccination may become positive or negative depending on whether features for old age, infancy, or immunodeficiency are included.

Transparency 3: Algorithmic Transparency

- Algorithmic transparency refers to understanding the **learning procedure itself**: will it converge? Does it produce a unique solution?
- For linear models, we often understand the error surface and can prove convergence to a unique solution.
- Deep learning usually lacks these kinds of theoretical guarantees.

Example

A linear regression model minimizes a convex loss surface, so we can **mathematically prove** it will always find the same best solution regardless of where training starts.

A deep neural network uses gradient descent on a non-convex surface full of local minima and saddle points. Training may converge to different solutions on different runs, and there is **no guarantee** it finds the global optimum, yet it works well in practice.

Post-hoc Interpretability 1: Text Explanations

- Post-hoc interpretations may provide useful information without exposing the true mechanism of the model.
- One form is a text explanation generated alongside or after the prediction.
- The paper warns these explanations may sound **plausible without being faithful** to the model's actual reasoning.

Example

Image Captioning as Explanation: a convolutional neural network first classifies an image; a second model then reads the internal representations of the first model and generates a natural-language caption describing what was seen. The caption acts as a human-readable explanation of the classification decision.

Recommender System with Review Text: a recommender system predicts star ratings using hidden latent factors; at the same time, a topic model reads user review text and links each hidden factor to interpretable topics. This makes the recommendation understandable to users by explaining not just what was recommended, but why.

Post-hoc Interpretability 2: Visualization

- Another common method is to visualize what the model has learned.
- Visualizations can help people inspect learned representations qualitatively.
- But they still do not fully reveal the complete mechanism of the model.

Example

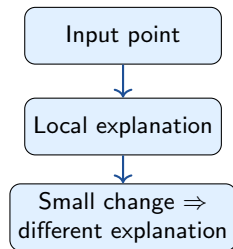
t-SNE is cited as a way to visualize high-dimensional representations in 2D.

Mordvintsev et al. (2015) enhanced hidden-unit activations and observed dog-face-like patterns appearing in images of the perturbed inputs.

Mahendran & Vedaldi (2015) showed that images could be reconstructed with high fidelity even from fairly high-level CNN representations (level 6 of AlexNet).

Post-hoc Interpretability 3: Local Explanations

- Local explanations describe what the model depends on near one particular input.
- They are useful, but **only locally valid**.
- The paper warns that local explanations can be misleading if treated as global understanding – moving a single pixel can produce a very different saliency map.



Example

saliency maps use gradients w.r.t. the input to highlight regions that most influence a prediction.

LIME: explains any model locally by fitting a sparse linear model in the neighborhood of a prediction point.

Post-hoc Interpretability 4: Explanation by Example

- Another way to explain a prediction is by showing the most similar examples.
- This is natural because humans also justify decisions by analogy.
- In learned representation spaces, nearest neighbors in the latent space can support this kind of explanation.

Example

Caruana et al. (1999) propose case-based explanation of non-case-based learning methods.

Doctors referring to similar case studies is the paper's human analogy.

Mikolov et al. (2013) inspect word2vec by checking nearest neighbors in latent space to understand what relationships the model has learned.

Linear Models vs. Deep Neural Networks

- Linear models are **not automatically more interpretable** than deep networks.
- Linear models may offer better **algorithmic transparency** due to well-understood optimization.
- With high-dimensional inputs or heavy feature engineering, linear models lose **simulatability** and **decomposability**.
- Deep networks often operate on raw or lightly processed features, keeping individual inputs more intuitive.
- Rich post-hoc analysis is possible on deep networks via visualization and representation inspection.
- Choosing between linear and deep models often trades **algorithmic transparency** against **decomposability**.

Example

Linear models approached RNN performance on clinical time series only through heavy feature engineering, which reduced decomposability and made individual features harder to interpret.

Claims must be qualified

- Every interpretability claim must fix a **specific definition**.
- Evidence must be provided that the method actually achieves its stated objective.

Post-hoc interpretations can mislead

- Explanations may be **plausible but unfaithful** to the model's actual reasoning.
- Mirrors how humans justify biased decisions with acceptable-sounding language.

Transparency can conflict with AI goals

- Transparent models built for doctor trust may sacrifice predictive performance.
- Interpretability should not be demanded unless it is **genuinely justified**.

Contestability is missing

- Published interpretability work rarely addresses **contestability**.
- Contestability is a person's right to meaningfully challenge an automated decision.
- This is central to the **EU right to explanation**.

- “Interpretability” is not one thing . it is a **collection of distinct concepts**.
- Any claim about interpretability must specify:
 - ▶ what kind of interpretability is meant (transparency vs. post-hoc),
 - ▶ and what real-world objective it is supposed to serve (trust, causality, fairness, etc.).
- Post-hoc explanations can help, but they can also **mislead**.
- Linear models are **not strictly more interpretable** than deep networks, it depends on which notion of interpretability is applied.
- Better **problem formulation** is as important as better algorithms.
- As machine learning continues to exert influence on society, the responsibility to ensure alignment with societal values must be shared by both **researchers and practitioners**.